

Md. Rakibul Hasan

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ORCID HuggingFace Kaggle PyPI

Professional Summary

Lead Engineer with 6+ years across software engineering, distributed data systems, and big data platforms — now specializing in enterprise GenAI architecture. Architecting LLMops pipelines, RAG knowledge retrieval systems, and autonomous AI agent frameworks at Banglalink Digital for the past 2 years. Hands-on in open-source LLM deployment, QLoRA fine-tuning (LoRA/PEFT), and GPU-based WebGPU inference. Active open-source contributor (PyPI, HuggingFace, 200+ GitHub repos); technical mentor across AI excellence initiatives. Undergraduate research in Transfer Learning.

Experience

Lead Engineer, Enterprise Data Engineering: Banglalink Digital, Dhaka, BD

- Sole data platform engineer and DevOps owner across full stack: IPDR billions-of-rows ETL and near-real-time microbatch pipeline for unplanned volume spikes, enterprise analytics on Kubernetes, and Airflow orchestration — 6 years continuous infrastructure ownership (Kubernetes, Docker Swarm, Helm, CI/CD, n8n, CCE)
- Architected enterprise **GenAI agent orchestration** platform (IT Toolkit: Tauri 2/Rust + MCP client/server) with multi-agent tool-calling, **autonomous workflow agents**, **context/memory management**, and **AI observability**; profiled for Slack, Jira, ServiceNow, Okta, M365
- Co-architected enterprise **RAG pipeline** (LangChain, LlamaIndex) for internal **knowledge retrieval**; designed **semantic search with embeddings**, **hallucination mitigation**, vector store integration, and **AI guardrails**; mentored GenAI engineering team
- Built **LLMops** pipeline for Day-2 Ops: autonomous **AI observability**, **model evaluation** hooks, LLM-powered L1 engineering assistant with **prompt management** and self-healing ETL
- Designed secure **Agent Sandbox** — AI agent evaluation environment with **responsible AI** controls, **AI governance** guardrails, and live browser/CLI access for agentic testing
- Architected AI-ready **distributed data platform** (Data Agent Ready Warehouse on S3) with **vector database**-compatible layer for **semantic search** and embedding workloads
- Built **Migration Hub**: GenAI-powered web UI with **LLM-assisted SQL translation** and **prompt engineering** for Teradata → Huawei DWS migration at scale
- Deployed enterprise self-service analytics platform (Apache Superset fork) on Kubernetes (solo DevOps, 100+ users); built **superset-hetuengine-connector** (open-source, PyPI — first-of-kind Superset → Huawei HetuEngine connector)
- Solo migrated tier-1 service Airflow 2.6 → 3.1.8 (zero unplanned downtime); architected IPDR billions-of-rows ETL and near-real-time billion-record pipeline for unplanned volume spikes
- Mentored ML/GenAI engineers; self-hosted n8n automation; built macOS VPN client (tigerstripes-vpn); created *yet-another-course* Python training program

Big Data Service Lead Engineer: Banglalink Digital, Dhaka, BD

- Deployed ML models (XGBoost) on CDP Cluster with **MLOps** practices; resolved source-level integration issues; engineered SIM lifecycle system (Spring Boot + MySQL)
- Developed PySpark **distributed** workflows; modernized network transmission with graph algorithms; built Rust/React desktop app for CSV validation
- Built real-time streaming pipeline with **Kafka** and Apache Ignite for high-throughput event processing on the big data platform

Software Engineer: Trucklagbe, Dhaka, BD

- Built API Gateway (Express/Nginx, 20+ microservices); Selenium Grid CI/CD (60% QA reduction); MITM Proxy inspector; business automation (Node.js, Google Apps Script)
- Analysed MongoDB mobile app user behaviour — categorised user actions, surfaced usage patterns, and presented findings via Metabase dashboards to inform product decisions

Junior Software Developer: Trucklagbe, Dhaka, BD

- GPS parser integrating 1500+ active devices; SaaS REST APIs for enterprise clients; cash disbursement module (Node.js, MySQL, Angular)

Software Engineering Consultant: Freelance / Vendor, Remote

- Delivered production platforms: c-onelogistics.com, programmerslab.tech, and others
- Built real-time Firebase/JS chat application for a Bangladeshi delivery service — coordinating riders, admins, and customers across live order flows
- Automated Statefarm.com web workflows using Java and Selenium — streamlined repetitive processes as a vendor engagement

Key Projects

Enterprise GenAI Agent Orchestration Platform (IT Toolkit): Tauri 2, Rust, Next.js 15, MCP, Fluent UI v9

Multi-agent **agent orchestration** (MCP client/server), CLI/browser/computer-use tool hierarchy, v2 **autonomous workflow agents** engine (auto/agent/human actors), **context/memory management**, **AI guardrails**, **AI observability**; enterprise site profiles (Slack, GitHub, Jira, ServiceNow, Okta)

Personal AI Agent — QLoRA Fine-tuned LLM: Python, PEFT, TRL, BitsAndBytes, HuggingFace

End-to-end **LoRA/PEFT** fine-tuning: TinyLlama 1.1B, 4-bit QLoRA (BitsAndBytes), LoRA rank-32, SFTTrainer, custom dataset curation; published as pesnik/pesnik-lora + pesnik/pesnik on HuggingFace; live **GPU-based inference** via WebGPU at pesnik.dev

Enterprise RAG Knowledge Retrieval System: LangChain, LlamaIndex, Python, Vector Store

Enterprise **RAG pipeline**: **semantic search**, **embeddings**, vector database integration, **hallucination mitigation**, **AI guardrails**, and context/memory management for Banglalink internal **knowledge retrieval**

LLMops & AI Observability Platform (DAGnostics): Python, FastAPI, LLMs, Drain3

Production **LLMops** monitoring: LLM-based ETL failure analysis, Drain3 log clustering, anomaly detection, **AI observability** dashboards, **model evaluation** and alerting pipeline

Migration Hub & tpt-builder — Teradata to Huawei DWS: TypeScript, Python, LLM, PyPI

GenAI-powered **prompt engineering** platform for LLM-assisted SQL dialect translation and query validation; **tpt-builder** (published on PyPI) generates Teradata PT scripts programmatically — core open-source tooling for enterprise DWH migration

Informatica-to-Airflow DAG Converter: TypeScript, Oracle

VS Code extension automating ETL migration; adopted by multiple engineers, saving 3+ hours per workflow

DAGSonar — Airflow Task Change Visibility: Python, Apache Airflow, PyPI

Tracks DAG modifications, version diffs, and configuration changes across Airflow pipelines; delivers real-time notifications to engineering teams; published as dagsonar on PyPI

PND Service BD — B2B/B2C E-commerce Platform: Laravel, MySQL, PHP, Docker

Full-stack e-commerce platform serving 6,000+ active users; payment gateway integration, inventory management, and order tracking; deployed on DigitalOcean with Docker

cocoon — Secure Password Manager: Rust

Memory-safe local password vault built in Rust; encrypted storage with zero external dependencies; demonstrates security-first systems design and Rust ownership model

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, Rust, Golang, SQL, C++, PHP, Shell, Swift

GenAI / LLMops: LLMops, RAG Pipelines, Agent Orchestration, Prompt Engineering, Semantic Search, Vector Databases, Embeddings, AI Guardrails, AI Observability, Model Evaluation, Hallucination Mitigation, Responsible AI, AI Governance

AI / ML: LangChain, LlamaIndex, LoRA/PEFT (QLoRA), Open-source LLM Deployment, WebGPU Inference, Fine-tuning, XGBoost, MLOps, MLflow, Anomaly Detection, MCP

Big Data / ETL: Apache Airflow, Spark, PySpark, Teradata, Informatica, Kafka, Huawei HetuEngine / DWS / MRS

Backend / DB: Spring Boot, Node.js, Flask, FastAPI, Express.js, Laravel, MySQL, Oracle, MongoDB, Redis

DevOps / Cloud: Kubernetes, Docker Swarm, Tauri 2, n8n, Huawei CCE, Helm, CI/CD, Apache Superset, Metabase

Education

BSc Computer Science & Engineering: University of Asia Pacific, CGPA: 3.77/4.00

Undergraduate research: **Transfer Learning** (model adaptation and fine-tuning)

Open Source & Research

HuggingFace: pesnik/pesnik (1B fine-tuned model, 40 downloads) · pesnik/pesnik-lora (QLoRA adapter, 26 downloads) · live **GPU-based WebGPU inference** at pesnik.dev

PyPI: 4 published packages (pypi.org/user/pesnik): **superset-hetuengine-connector** (Superset → Huawei HetuEngine, first-of-kind) · **dagsonar** (Airflow task change visibility) · **diagnostics** (LLM ETL monitoring) · **tpt-builder** (Teradata PT script builder)

GitHub: 200+ public repos (github.com/pesnik); open PRs to **Daytona** (AI dev sandbox) and **Excalidraw**; DAGnostics (LLM ETL monitoring)

Research: ORCID: 0009-0006-4082-5527 · Kaggle: pesnik

Awards & Achievements

Professional: Spot Award for AI solutions, CTIO Excellence Award (Airflow migration leadership), Game Changer Award (2×) — Banglalink Digital

Academic: Dean's Award (4×), Vice Chancellor's Award, 1st Place UAP Programming Contest, 1st Place National Essay Writing Competition (Red Cross BD), ACM ICPC Regional 36th